Haitao(Book) Chen
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05/27/2026

A. Education

- **Ph.D.** 2023 - present Industrial and Manufacturing Engineering, Human Factors
  Pennsylvania State University, State College, PA
- **B.S.** 2018 - 2022 Industrial Engineering
  Tianjin University, Tianjin, China

B. Research Interests

- Driver Cognition and Behavior in Mixed Traffic
- Trust in Automation
- Situation Awareness
- Human-Machine Interface Design

C. Research Project

1. CSRAI Seed Grant, Human-Centered Artificial Intelligence for Public Transportation Services, Penn State, $ 50,000, 2026-Present. PI: Aron Laska.
    - Study team member
    - Develop protocol for Human Subject Research
    - Data analysis and management
2. Big Ideas Grant, Enhancing Situation Awareness of Adversary ML in Human-AI Collaboration, Penn State, $ 50,000, 2023-Present. PI: Yiqi Zhang.
    - Study team member
    - Experimental design, conduct experiments in driving simulator, data analysis.

D. Publications

Peer-reviewed Journal Papers

1. **Chen, H**., & Zhang, Y. (2026). Mitigating driver confusion and aggression in mixed traffic: Effects of V2V communication and driving style. *Accident Analysis & Prevention*, 236, 108676.
2. **Chen, H.,** Ma, Z., & Zhang, Y.*. (2026). Behavioral adaptation in mixed traffic: the roles of AV penetration rate and human driving style. *Accident Analysis & Prevention*, 232, 108516.
3. **Chen, H.**, Zhang, Y.*. (2026). Mitigating aggression toward AVs in mixed traffic: Effects of automated driving styles and human driving styles. *Transportation Research Part F: Traffic Psychology and Behaviour*, *116*, 103431.

Manuscripts under Review and Preparation

1. **Chen, H.**, Zhang, Y*. (In preparation). Aggression toward automated vehicles drives human takeover behavior.

2. **Chen, H**., Gao, J., Xiong, A., Zhang, Y.* (In preparation). Under Adversarial Attack, the effects of SA-Based Augmented Reality HUDs on Driver Situation Awareness in Level 3 ADS.

3. **Chen, H.**, Wu, Z., Hu, X., Zhang, Y.* (In preparation). Characterizing Decision-Making Differences of Automated Vehicles and Human-Driven Vehicles at Unsignalized Intersections.

Peer-reviewed Conference Proceeding Papers

In the below, a # indicates the presenting author.

1. **Chen, B.** #, Zhang, Y.*. (2026). Mitigating driver confusion and aggressive driving behavior in mixed traffic using V2V communication. Transportation Research Board Annual Meeting.

2. **Chen, B.#,** Zhang, Y.*. (2026). When Automated Vehicles Cannot Retaliate: Driver-Initiated Takeover Under Aggression in Mixed Traffic. Proceedings of the Human Factors and Ergonomics Society Annual Meeting

3. **Chen, B.#**, Gao, J., Xiong, A., Zhang, Y.*. Chen, H., Gao, J., Xiong, A., Zhang, Y.* (In preparation). Under Adversarial Attack, the effects of SA-Based Augmented Reality HUDs on Driver Situation Awareness in Level 3 ADS. Proceedings of the Human Factors and Ergonomics Society Annual Meeting

4. **Chen, B.** #, Zhang, Y.*. (2025). Mitigating aggressive driving in mixed traffic: The role of driving styles of automated vehicles and human-driven vehicles. Transportation Research Board Annual Meeting.

E. Teaching

• **IE 327 Introduction to Work Design**	Fall 2026
Instructor	Pennsylvania State University
• **IE 327 Introduction to Work Design**	Spring 2024, Spring 2025, Fall 2025
Teaching Assistant	Pennsylvania State University

F. Undergraduate Student Advising

1. Xinlu Huang, Enhancing Situation Awareness of Adversary ML in Human-AI Collaboration, 2025- Present.
2. Alexander Maldonado, Enhancing Situation Awareness of Adversary ML in Human-AI Collaboration, 2025- Present.
3. Samir Jacobo Ramos, “The effect of HMI design of HV-AV interaction in mixed traffic.” January 2025- present.
4. Connor Legge, “The effect of automated driving styles on drivers’ decision-making in mixed traffic.” January 2024- May 2024.
5. Xinyi Zhou, “The effect of automated driving styles on drivers’ decision-making in mixed traffic.” January 2024- May 2024.

G. Review Experience

- Reviewer, Transportation Research Part F: Traffic Psychology and Behaviour.

- Reviewer, Transactions on Haptics.
- Reviewer, IEEE Transactions on Intelligent Transportation Systems.
- Reviewer, Advanced Engineering Informatics.
- Reviewer, Automotive UI' 2025.
- Reviewer, Proceedings of the Human Factors and Ergonomics Society.

H. Professional Leadership, Outreach and Service

Professional Affiliation

1. Member, Transportation Research Board, 2024- Present
2. Member, Human Factors and Ergonomics Society, 2023-Present.

I. Awards

1. **Marcus-Niebel-Schall Graduate Student Scholarship,** $ 10,000, Penn State Department of Industrial and Manufacturing Engineering, 2026 (*Awarded annually to 5 Ph.D. students who demonstrate excellence in both academic performance and research*).
2. **Harry G. Miller Fellowships in Engineering,** $12,000**,** Penn State College of Engineering, 2025 (*Awarded annually to outstanding engineering students*).
3. **Penn State Graduate Student International Travel Grant**, $1,500, Penn State University, 2025 (*Competitive international travel award from Pennsylvania State University to present research at an international conference*).